

MANGA AUTOMATION SYSTEM

Technical Implementation Guide

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Brainrot content pipeline	Product ad pipeline
n8n + Mastra + ElevenLabs + short-video-maker + Kling AI	Firecrawl + Claude + Creatomate + TikTok Shop affiliate

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1. System architecture overview

The system is composed of two independent but interconnected pipelines sharing the same infrastructure layer. The brainrot lore pipeline runs daily (or more) and drives traffic. The product ad pipeline runs 2–3 times per week and monetises that traffic.

1.1 High-level architecture

Intelligence layer	Production layer	Distribution layer
• Mastra AI agent • Claude API (script gen) • Perplexity (trend data) • n8n orchestration	• ElevenLabs (TTS) • Kling AI (panel anim) • short-video-maker • Creatomate (product ads)	• TikTokAutoUploader • YouTube Data API • Instagram Graph API • Pinterest API

Design principle

Every pipeline step is stateless and idempotent. If any step fails, n8n retries it independently. State (chapter data, video status, analytics) is stored in a Postgres database, not in workflow memory.

1.2 Data flow

The system processes one manga chapter at a time through the following sequence:

→	Firecrawl detects new chapter on MangaDex/Webtoons → stores raw text + panel URLs in Postgres
→	n8n trigger fires → passes chapter payload to Mastra agent
→	Mastra generates 3 script variants → Claude scores them → winning script selected
→	ElevenLabs converts script to MP3 with word timestamps → stored in S3/R2
→	Kling AI animates top 3 panels → 2–3 second clips → stored in S3/R2
→	short-video-maker assembles final MP4: animated panels (top) + gameplay (bottom) + synced captions
→	Creatomate appends 5-second product CTA overlay → final MP4
→	TikTokAutoUploader posts video → YouTube/Instagram APIs repost 30 mins later
→	24h post: analytics fetched → performance logged → Mastra memory updated

2. Infrastructure & prerequisites

2.1 Server requirements

OS	Ubuntu 22.04 LTS (recommended) or macOS 14+
RAM	Minimum 4GB for n8n + short-video-maker. 8GB recommended for concurrent renders
Storage	50GB+ for video asset cache. Use S3/Cloudflare R2 for produced MP4 files
Docker	Docker Engine 24+ and Docker Compose v2 required for short-video-maker
Node.js	v20+ (required for n8n self-hosted and short-video-maker)
Python	3.11+ (for any custom scraping scripts and Mastra helper utilities)

2.2 Required API keys

Collect all of these before beginning setup. Store them in a .env file and reference them via n8n's built-in credential store — never hardcode in workflow nodes.

Service	Where to get it	Used for
Anthropic (Claude)	<code>console.anthropic.com</code>	Script generation, scoring, analytics summary
ElevenLabs	<code>elevenlabs.io/developers</code>	Text-to-speech voiceover
Kling AI	<code>klingai.com/api</code>	Manga panel animation
Creatomate	<code>creatomate.com</code>	Product ad video rendering
Firecrawl	<code>firecrawl.dev</code>	Anti-bot-proof web scraping
MangaDex API	<code>api.mangadex.org</code> (free)	Chapter detection and panel URLs
Pexels API	<code>pexels.com/api</code> (free)	Gameplay background footage
TikTok API	<code>developers.tiktok.com</code>	Video upload + analytics
YouTube Data API	<code>console.cloud.google.com</code>	Shorts + long-form upload
Instagram Graph API	<code>developers.facebook.com</code>	Reels upload
Cloudflare R2 / AWS S3	<code>cloudflare.com</code> or <code>aws.amazon.com</code>	Video and audio asset storage
Perplexity API	<code>perplexity.ai/api</code>	Trending topic research

2.3 Database setup (Postgres)

```
-- Core tables required by the system
CREATE TABLE chapters (
  id          SERIAL PRIMARY KEY,
  manga_id    TEXT NOT NULL,
  chapter_num TEXT NOT NULL,
  title       TEXT,
  panel_urls  JSONB,
  raw_text    TEXT,
  scraped_at  TIMESTAMPTZ DEFAULT NOW(),
  processed   BOOLEAN DEFAULT FALSE
);

CREATE TABLE videos (
  id          SERIAL PRIMARY KEY,
  chapter_id  INT REFERENCES chapters(id),
  script_text TEXT,
  audio_url   TEXT,
  video_url   TEXT,
  tiktok_id   TEXT,
  youtube_id  TEXT,
  ig_id       TEXT,
  posted_at   TIMESTAMPTZ,
  views_24h   INT,
  completion_rt NUMERIC
);

CREATE TABLE products (
  id          SERIAL PRIMARY KEY,
  shop_id     TEXT,
  name        TEXT,
  price       NUMERIC,
  image_url   TEXT,
  commission_pct NUMERIC,
  affiliate_url TEXT,
  last_scraped TIMESTAMPTZ
);
```

3. Tool stack — full breakdown

3.1 Orchestration — n8n

n8n (self-hosted) Open source — Free

Visual workflow automation with 400+ integrations. The backbone connecting every tool in the system.

- Install via Docker: `docker run -d -p 5678:5678 -v n8n_data:/home/node/.n8n docker.n8n.io/n8nio/n8n`
- Access at `http://localhost:5678` — create your first workflow via the visual canvas
- Use built-in Credentials store for all API keys — never hardcode in nodes
- Enable webhook triggers for real-time events (TikTok upload confirmation, MangaDex webhooks)
- Set up error workflows to Slack/Discord so failures notify you immediately

Tip: n8n's HTTP Request node gets blocked by modern anti-bot systems. Always route scraping through Firecrawl — not directly through n8n's built-in request node.

```
# Docker Compose setup for n8n + Postgres
version: '3.8'
services:
  n8n:
    image: docker.n8n.io/n8nio/n8n
    ports: ['5678:5678']
    environment:
      - DB_TYPE=postgresdb
      - DB_POSTGRESDB_HOST=postgres
      - DB_POSTGRESDB_DATABASE=n8n
      - DB_POSTGRESDB_USER=n8n
      - DB_POSTGRESDB_PASSWORD=${POSTGRES_PASSWORD}
    volumes: ['n8n_data:/home/node/.n8n']
  postgres:
    image: postgres:16
    environment:
      POSTGRES_DB: n8n
      POSTGRES_USER: n8n
      POSTGRES_PASSWORD: ${POSTGRES_PASSWORD}
    volumes: ['postgres_data:/var/lib/postgresql/data']
```

3.2 AI agent — Mastra

Mastra Open source — Free

TypeScript-first AI agent framework. Handles the LLM reasoning layer: reading chapters, generating scripts, scoring variants, and learning from analytics.

- Install: `npm create mastra@latest` — scaffolds agent project with TypeScript
- Mastra handles tool calling, memory, and multi-step reasoning natively
- Connect to Claude via Anthropic SDK inside the agent's tool definitions
- Agents expose HTTP endpoints that n8n calls as standard REST API nodes

- Use Mastra's built-in memory to store which script styles performed best per manga series

Tip: Keep Mastra agents focused on LLM reasoning only. All I/O (file storage, API calls, scheduling) belongs in n8n — not in the agent code.

```
// mastra/agents/manga-scriptwriter.ts
import { Agent } from '@mastra/core';
import Anthropic from '@anthropic-ai/sdk';

export const scriptwriterAgent = new Agent({
  name: 'MangaScriptwriter',
  instructions: `You are a viral manga content creator.
  Given a manga chapter summary, generate 3 short-form video scripts.
  Each script must be under 60 seconds when read aloud.
  Format: {hook, body, cta} — always start with a controversial hook.
  Scoring criteria: controversy > novelty > educational value.`,
  model: {
    provider: 'ANTHROPIC',
    name: 'claude-sonnet-4-6',
  },
  tools: { scoreScript, fetchTrendingTopics, logPerformance },
});
```

3.3 Scraping — Firecrawl

Firecrawl Paid — from \$16/mo

Managed scraping API that bypasses Cloudflare, JavaScript rendering, and all modern anti-bot systems. Essential — raw HTTP requests are blocked by MangaDex and TikTok Shop.

- API endpoint: <https://api.firecrawl.dev/v1/scrape> — pass URL + extraction schema
- Use 'extract' mode with a JSON schema to get structured chapter data directly (no HTML parsing)
- Supports markdown output, screenshot capture, and JavaScript-rendered content
- Budget: ~\$16–49/month covers your scraping volume at 1–2 requests per chapter

Tip: Use Firecrawl's 'extract' mode with a defined schema instead of scraping HTML and parsing manually. It's faster and doesn't break when sites change their layout.

```
// n8n HTTP Request node — Firecrawl extract
POST https://api.firecrawl.dev/v1/extract
Authorization: Bearer ${FIRECRAWL_API_KEY}

{
  "urls": ["https://mangadex.org/chapter/${CHAPTER_ID}"],
  "prompt": "Extract the chapter title, chapter number, panel image URLs, and any dialogue text",
  "schema": {
    "type": "object",
    "properties": {
      "chapter_title": {"type": "string"},
      "chapter_number": {"type": "string"},
      "panel_urls": {"type": "array", "items": {"type": "string"}},
    }
  }
}
```

```

    "dialogue": {"type": "array", "items": {"type": "string"}}
  }
}

```

3.4 TTS — ElevenLabs

ElevenLabs API Paid — from \$22/mo

Best-in-class expressive text-to-speech with emotion tags, 32 languages, 10K+ voices, and word-level timestamps for caption sync.

- Model: eleven_multilingual_v2 for highest quality; eleven_flash_v2_5 for real-time (75ms latency)
- Request word-level timestamps by setting timestamp_granularities=['word'] — required for caption sync
- Clone one consistent narrator voice and reuse its voice_id across all videos for channel branding
- Use emotion tags in scripts: [excited], [whispers], [laughs] — ElevenLabs v3 respects them
- Cost: 1 credit per character. A 60-second script (~750 characters) costs ~750 credits (~\$0.02)

Tip: Always test new scripts with Kokoro-js (free, local) before spending ElevenLabs credits. Only render final approved scripts through ElevenLabs.

```

// ElevenLabs API call with word timestamps
const response = await fetch(
  `https://api.elevenlabs.io/v1/text-to-speech/${VOICE_ID}/with-timestamps`,
  {
    method: 'POST',
    headers: {
      'xi-api-key': process.env.ELEVENLABS_API_KEY,
      'Content-Type': 'application/json',
    },
    body: JSON.stringify({
      text: scriptText,
      model_id: 'eleven_multilingual_v2',
      voice_settings: { stability: 0.5, similarity_boost: 0.75 },
    }),
  }
);

const { audio_base64, alignment } = await response.json();
// alignment.characters = word array with start/end timestamps
// Use these for caption sync in short-video-maker

```

3.5 Video assembly — short-video-maker

short-video-maker Open source — Free

Docker container exposing a REST + MCP API. Combines TTS, auto-captions, background footage, and music into a finished vertical MP4. No GPU required.

- Run: `docker run -d -p 3123:3123 -e PEXELS_API_KEY=xxx gyoridavid/short-video-maker:latest-tiny`
- POST `/api/short-video` with scenes array — each scene has text + searchTerm for Pexels footage
- Built on Remotion under the hood — outputs 9:16 vertical MP4 ready for TikTok/Shorts/Reels
- Captions rendered word-by-word from ElevenLabs timestamps — set `captionPosition`: 'top' for split-screen
- For split-screen: pass your animated manga panels as the `videoUrl` instead of Pexels search results

Tip: The 'tiny' Docker image runs on 2GB RAM. For concurrent renders, use 'latest' image on a 4GB+ server or run multiple containers behind a simple load balancer.

```
// POST http://localhost:3123/api/short-video
{
  "scenes": [
    {
      "text": "[excited] Sukuna just destroyed an ENTIRE mountain",
      "videoUrl": "https://your-r2.workers.dev/panel-animated-01.mp4",
      "durationMs": 3500
    },
    {
      "text": "and fans think THIS is his true power level?",
      "searchTerm": "subway surfers gameplay", // fallback: Pexels
      "durationMs": 4000
    }
  ],
  "voiceoverUrl": "https://your-r2.workers.dev/audio-script-001.mp3",
  "captionPosition": "top",
  "captionBackgroundColor": "#FFDD00",
  "orientation": "portrait",
  "musicTag": "energetic",
  "musicVolume": "low"
}
```

3.6 Panel animation — Kling AI 2.1

Kling AI 2.1 API **Paid — per clip**

State-of-the-art image-to-video model. Animates static manga panels into 2–5 second cinematic clips with zoom, pan, and particle motion.

- API endpoint: `https://api.klingai.com/v1/videos/image2video`
- Input: panel image URL + motion prompt. Output: MP4 clip URL (polling or webhook)
- Use motion prompts like: 'slow cinematic zoom in, dramatic lighting pulse, dust particles'
- Generate 3–4 animated clips per video — one per key manga moment
- Cost: approximately \$0.05–0.15 per 5-second clip depending on resolution

Tip: Keep animation prompts short (under 30 words) and motion-focused. Avoid describing the image content — Kling already sees it. Just describe the camera and atmosphere.

3.7 Product ad rendering — Creatomate

Creatomate Paid — from \$29/mo

Template-based cloud video rendering API. Build your TikTok Shop ad template once in their visual editor, then generate unlimited product ad videos via JSON API.

- Build 3 templates in Creatomate editor: (1) product reveal, (2) before/after, (3) testimonial-style
- Each template has dynamic fields: {{product_image}}, {{product_name}}, {{price}}, {{cta_text}}
- API endpoint: POST <https://api.creatomate.com/v1/renderers>
- Render time: 15–45 seconds per video. Webhook notifies n8n when MP4 is ready
- Output: direct MP4 URL ready to upload — no intermediate download step needed

Tip: Create a 'dark mode' and 'light mode' version of each template. Alternate between them to avoid ad fatigue — same product, different visual feel.

```
// POST https://api.creatomate.com/v1/renderers
// Authorization: Bearer ${CREATOMATE_API_KEY}
{
  "template_id": "your-tiktok-shop-ad-template-id",
  "modifications": {
    "product_image": "https://cdn.tiktokshop.com/product/gojo-eyepatch.jpg",
    "product_name": "Gojo Satoru Infinity Eyepatch",
    "price": "$15.99",
    "cta_text": "Perfect for JJK fans — tap to shop",
    "background_color": "#0D0D0D",
    "accent_color": "#1A56A0"
  },
  "webhook_url": "https://your-n8n.app/webhook/creatomate-done"
}
```

4. Brainrot lore video pipeline

The complete step-by-step implementation of the daily content pipeline. Each step corresponds to one or more n8n nodes.

4.1 Step 1 — Chapter detection

Trigger

n8n Cron node fires every 6 hours. Uses MangaDex API (free, no scraping needed for chapter metadata) to check for new chapters. Falls back to Firecrawl for sites without a public API (Webtoons, VizMedia).

```
// MangaDex API — get latest chapters for a manga
GET https://api.mangadex.org/manga/{MANGA_ID}/feed
  ?limit=5
  &order[chapter]=desc
  &translatedLanguage[]=en

// Response contains chapter IDs, numbers, and page image URLs
// Compare against chapters table — if chapter_id not present, it's new

// n8n IF node: chapters.find(c => c.id === data.chapterId) === undefined
// If true → proceed to Step 2. If false → stop (already processed).
```

4.2 Step 2 — Script generation

```
// n8n HTTP Request → Mastra agent endpoint
POST http://mastra-server:4000/agents/manga-scriptwriter/generate
{
  "chapter_text": "{{ $node.firecrawl.json.dialogue.join(' ') }}",
  "chapter_title": "{{ $node.mangadex.json.data.attributes.title }}",
  "manga_series": "Jujutsu Kaisen",
  "trending_keywords": "{{ $node.perplexity.json.keywords }}",
  "previous_performance": "{{ $node.postgres.json.best_script_style }}"
}

// Mastra returns 3 scripts in format:
// { scripts: [ {hook, body, cta, style, score}, ... ] }
// n8n selects the highest scoring script automatically
```

Script format — what Claude generates

Every script follows the structure: [HOOK — controversial claim, <5 seconds] → [BODY — 3–5 rapid-fire points, 35–45 seconds] → [CTA — TikTok Shop product mention, 5–8 seconds]. Target: 50–60 seconds total read time at ElevenLabs default speed.

4.3 Step 3 — Voice generation

```
// n8n HTTP Request → ElevenLabs
POST https://api.elevenlabs.io/v1/text-to-speech/{VOICE_ID}/with-timestamps
xi-api-key: ${ELEVENLABS_API_KEY}

{
  "text": "{{ $node.mastra.json.scripts[0].hook }} {{
$node.mastra.json.scripts[0].body }}",
  "model_id": "eleven_multilingual_v2",
  "voice_settings": {"stability": 0.45, "similarity_boost": 0.8}
}

// Response: { audio_base64, alignment: { characters, start_times, end_times } }
// Decode base64 → upload MP3 to R2 → store audio_url + alignment in Postgres
videos table
```

4.4 Step 4 — Panel animation (Kling AI)

```
// n8n HTTP Request → Kling AI (per panel, run as 3 parallel branches)
POST https://api.klingai.com/v1/videos/image2video
Authorization: Bearer ${KLING_API_KEY}

{
  "model_name": "kling-v2-master",
  "image_url": "{{ $node.firecrawl.json.panel_urls[0] }}",
  "prompt": "slow zoom in, dramatic shadows, atmospheric dust particles,
cinematic",
  "duration": "5",
  "aspect_ratio": "9:16"
}

// Response: { task_id } — poll GET /v1/videos/{task_id} until status = 'completed'
// n8n Wait node: retry every 15s for up to 3 minutes
// On completion: store clip URLs in array for Step 5
```

4.5 Step 5 — Video assembly

```
// Build scenes array for short-video-maker
// Top half: Kling-animated manga panels cycling
// Bottom half: gameplay footage (Pexels search or your own pool)

POST http://short-video-maker:3123/api/short-video
{
  "scenes": [
    { "text": "HOOK_TEXT",          "videoUrl": "{{ panel_clip_01 }}", "durationMs":
4000},
    { "text": "BODY_POINT_1",      "videoUrl": "{{ panel_clip_02 }}", "durationMs":
8000},
    { "text": "BODY_POINT_2",      "searchTerm": "parkour action",    "durationMs":
8000},
    { "text": "BODY_POINT_3",      "videoUrl": "{{ panel_clip_03 }}", "durationMs":
8000},
    { "text": "BODY_POINT_4",      "searchTerm": "speedrun gaming",   "durationMs":
8000},
  ]
}
```

```

    {"text": "CTA_TEXT",          "searchTerm": "anime merchandise", "durationMs":
7000}
  ],
  "voiceoverUrl": "{{ $node.r2.json.audioUrl }}",
  "captionPosition": "top",
  "captionBackgroundColor": "#FFDD00",
  "captionActiveTextColor": "#000000",
  "musicTag": "dramatic",
  "musicVolume": "low",
  "orientation": "portrait"
}

// Returns: { jobId } - poll GET /api/short-video/{jobId} for status
// On success: response.videoUrl = final MP4 URL

```

4.6 Step 6 — CTA overlay (Creatomate)

The base video from short-video-maker is 50–55 seconds. Creatomate appends a 5-second animated product card at the end, bringing total runtime to approximately 60 seconds — the minimum for Creator Rewards eligibility.

4.7 Step 7 — Upload

Upload sequence

TikTok first (primary platform). YouTube Shorts 30 minutes later (different time zone peak). Instagram Reels 60 minutes after YouTube. This staggered timing prevents the algorithm from treating them as coordinated simultaneous posts.

```

# TikTokAutoUploader - haziq-exe version
from tiktok_uploader import upload_video

upload_video(
    video_path='output_final.mp4',
    description=f'{hashtags}\n\n{product_cta}',
    cookies='cookies.json',
    # Sound: use a trending sound from TikTok's current chart
    # Caption: Claude-generated, 150 chars max, ends with 2-3 hashtags
)

```

5. Product ad pipeline

Runs independently of the lore pipeline. Identifies high-commission TikTok Shop products in the anime/manga category and generates polished 15-second product showcase videos.

1 Product discovery

Firecrawl scrapes TikTok Shop bestseller list filtered by: commission $\geq$ 10%, reviews $\geq$ 100, rating $\geq$ 4.0, category = anime/manga/cosplay. Returns structured product JSON to Postgres.

```
// Filter criteria in Firecrawl extract schema
"filter": {
  "commission_rate": {"minimum": 0.10},
  "review_count":    {"minimum": 100},
  "star_rating":     {"minimum": 4.0},
  "ships_from":      "local"
}
```

2 Script variants

Claude generates 3 hook angles for the product, each targeting a different viewer: (1) cosplay/fan identity angle, (2) gift idea angle, (3) quality/value angle. n8n stores all three and rotates through them across 3 separate ad posts.

3 Creatomate render

API call passes product_image, product_name, price, cta_text, and hook_text. Three template variants are rotated: product reveal (dark), before/after (light), social proof (minimal). Returns MP4 via webhook.

4 Scheduled upload

Product ads posted between organic lore videos — never back-to-back ads. Optimal ratio: 3 lore videos per 1 product ad. Tagged with #TikTokShop and affiliate product link.

Separation rule

Never embed TikTok Shop product links directly inside your lore videos. Keep them as separate dedicated ad videos. Algorithm detects when organic content is overloaded with shopping links and reduces distribution. The CTA overlay in lore videos should only reference your own Whop/Patreon or generic merch — save hard affiliate links for dedicated shop videos.

6. n8n workflow configuration

6.1 Workflow 1 — Chapter monitor (runs every 6 hours)

Node 1	Cron trigger — fires at 0, 6, 12, 18 every day
Node 2	HTTP Request → MangaDex API — get latest 5 chapters for each tracked manga
Node 3	Postgres — check if chapter_id already exists in chapters table
Node 4	IF node — if new chapter: continue. If exists: stop
Node 5	Firecrawl HTTP Request — scrape full chapter page for dialogue and panel URLs
Node 6	Postgres insert — save chapter data, set processed = FALSE
Node 7	HTTP Request → Mastra scriptwriter agent
Node 8	ElevenLabs TTS → store MP3 to R2
Node 9	Kling AI animate panels — 3 parallel branches, merge after all complete
Node 10	HTTP Request → short-video-maker → poll for completion
Node 11	Creatomate CTA overlay → webhook trigger on completion
Node 12	Wait 5 minutes
Node 13	Python script node → TikTokAutoUploader
Node 14	Wait 30 minutes → YouTube Shorts upload via YouTube Data API
Node 15	Wait 60 minutes → Instagram Reels upload via Graph API
Node 16	Postgres update — set processed = TRUE, store platform IDs
Node 17	Wait 24 hours — Cron or scheduled workflow
Node 18	Fetch analytics from TikTok/YouTube APIs
Node 19	Postgres update — store views_24h, completion_rt
Node 20	HTTP Request → Mastra performance logger — update script style weighting

6.2 Error handling strategy

Error workflow

Every workflow has an 'onError' path pointing to a dedicated error handler workflow. This workflow: (1) logs the error to Postgres, (2) sends a Slack/Discord notification with node name and error message,

(3) retries the failed step once after 5 minutes, (4) marks the chapter as 'error' in the database if retry also fails. This prevents silent failures from breaking your posting schedule.

```
// n8n error workflow trigger (Error Trigger node)
// Receives: { workflow, node, error, execution }

// Step 1: Log to Postgres
INSERT INTO errors (workflow_id, node_name, error_msg, chapter_id, occurred_at)
VALUES ($1, $2, $3, $4, NOW())

// Step 2: Slack notification
POST https://hooks.slack.com/services/YOUR_WEBHOOK
{
  "text": ":warning: n8n error in {{ $node.errorWorkflowName }}"
    "Node": {{ $node.lastNodeExecuted }}
    "Error": {{ $node.errorMessage }}"
}
```

6.3 Parallel execution limits

Chapter ingestion	Sequential only — one chapter at a time to avoid Kling AI rate limits
Panel animation	3 parallel Kling requests per chapter (one per panel)
Product ad renders	Up to 5 concurrent Creatomate renders (their API supports this)
Upload queue	Sequential with time delays — no parallel uploading to same platform
Analytics fetching	Can run in parallel — lightweight read-only API calls

7. Mastra agent setup

7.1 Agent project structure

```
manga-automation/
├── mastra/
│   ├── agents/
│   │   ├── scriptwriter.ts      # Main script generation agent
│   │   ├── product-researcher.ts # TikTok Shop product analysis
│   │   └── performance-logger.ts # Analytics → future script weighting
│   ├── tools/
│   │   ├── score-script.ts      # Ranks 3 scripts by engagement potential
│   │   ├── fetch-trending.ts    # Perplexity API for trending keywords
│   │   ├── log-performance.ts   # Updates Postgres performance weights
│   │   └── caption-generator.ts # TikTok caption + hashtag generation
│   └── index.ts                 # Express server exposing agent endpoints
└── docker-compose.yml
```

7.2 Script scoring logic

```
// tools/score-script.ts
// Scores each generated script variant 0-100

function scoreScript(script: Script, context: Context): number {
  let score = 0;

  // Hook controversy (0-40 pts)
  // Does the hook make a falsifiable, debatable claim?
  if (isControversialClaim(script.hook)) score += 40;
  else if (isQuestionHook(script.hook)) score += 20;

  // Trending keyword overlap (0-30 pts)
  const overlap = countKeywordOverlap(script.body, context.trendingKeywords);
  score += Math.min(overlap * 10, 30);

  // Historical performance of this style (0-30 pts)
  const historicalScore = context.styleWeights[script.style] || 15;
  score += historicalScore;

  return score;
}
```

7.3 Memory and learning loop

After each video's 24-hour analytics are collected, n8n calls the performance-logger agent with the video's script style, view count, and completion rate. The agent updates a `style_weights` table in Postgres. This weight table influences scoring in future script generation runs — the system learns which formats work for each manga series over time.

```
// performance-logger agent tool
async function logPerformance(input: {
```

```
script_style: string,          // 'controversy' | 'lore_reveal' | 'tier_list'
views_24h: number,
completion_rate: number,      // 0-1
manga_series: string
})) {
  const weight = (input.completion_rate * 0.6) + (Math.log(input.views_24h) * 0.4);

  await db.query(`
    INSERT INTO style_weights (manga_series, style, weight, updated_at)
    VALUES ($1, $2, $3, NOW())
    ON CONFLICT (manga_series, style)
    DO UPDATE SET weight = (style_weights.weight * 0.7) + ($3 * 0.3)
    `, [input.manga_series, input.script_style, weight]);
}
// Exponential moving average: 70% historical + 30% new data
```

8. Content formats & scripts

8.1 The 7 primary content formats

Your Mastra agent generates scripts in one of these 7 formats. Each has a different hook structure, body format, and optimal use case. Train your agent with these as examples in the system prompt.

Format 1 — Hot take / controversy bait

Hook formula	[Number] reasons why [beloved character] is actually [negative trait]
Example hook	3 reasons why Gojo Satoru is actually the most overrated character in JJK
Body structure	3–5 rapid points, each contradicting the fan consensus
Best for	JJK, One Piece, Naruto — large fandoms that argue passionately
Expected engagement	Highest comment count of all formats — algorithm gold

Format 2 — Hidden lore reveal

Hook formula	Most people missed THIS in chapter [N] of [series]
Example hook	Most people missed this detail in JJK chapter 261 — here's what it actually means
Body structure	Present the 'hidden' panel, explain the foreshadowing, reveal the implication
Best for	Lore-dense series: JJK, Berserk, Vinland Saga, Chainsaw Man
Expected engagement	Highest share rate — people send these to friends who read the series

Format 3 — Power level debate

Hook formula	Can [Character A] actually beat [Character B]? Here's the math
Example hook	Can Gojo beat Madara? I did the math and the answer will make you angry
Body structure	Present each character's stats, find the decisive factor, give a verdict
Best for	Cross-series comparisons — JJK vs Naruto, One Piece vs Dragon Ball
Expected engagement	Second-highest comment count — fans always disagree with the verdict

Format 4 — Manga vs anime comparison

Hook formula	The anime CHANGED this scene from the manga — here's what you missed
Example hook	MAPPA completely changed this JJK chapter and most anime-only fans don't know
Body structure	Show manga panel, show anime scene, explain the change and its significance
Best for	Any series with both manga and active anime adaptation
Expected engagement	Natural funnel — viewers who only watch anime get curious about the manga

Format 5 — Chapter summary speed-run

Hook formula	[Series] chapter [N] in 60 seconds — this is what happened
Example hook	JJK chapter 271 in 60 seconds — Gojo is back and everything just changed
Body structure	Chronological summary hitting only the 3–4 most significant moments
Best for	Every new chapter, posted within hours of release
Expected engagement	Highest search traffic — people search 'jjk chapter 271 explained'

8.2 Caption template

```
# n8n Function node — caption generator
// Claude generates captions following this exact structure:

const caption = `
${hookText.toUpperCase()} 🔥

${seriesName} fans you need to see this 📺

Follow for daily manga lore drops 📖

${productCta}

#${seriesHashtag} #manga #animefyp #${characterName} #lore #fyp
`.trim();

// Max 150 chars before hashtags. Product CTA: 'Shop our JJK collection 🛒 link in bio'
// Always end with exactly 5 hashtags — TikTok's optimal hashtag count
```

9. Platform distribution

9.1 Posting schedule

TikTok	3–5 videos/day. Peak times: 7–9am, 12–2pm, 7–10pm (audience local time). Use Creator Rewards eligible format: 60-second minimum, original content.
YouTube Shorts	2–3 videos/day. Post 30 mins after TikTok. Repurpose same MP4 — YouTube's algorithm treats it as original if not watermarked.
Instagram Reels	1–2 videos/day. Post 60 mins after YouTube. Focus on your highest-performing TikToks — don't repost everything.
Facebook Reels	1 video/day. Use lowest-performing TikTok content — older, different audience. Good for long-tail ad revenue.
Pinterest	3–5 pins/day. Use Flux-generated character art + chapter title. Drives Amazon affiliate traffic passively.
YouTube long-form	1 video/week. Compile 5 lore scripts into a 10–15 minute 'Full chapter breakdown' video. 5–8x higher CPM than Shorts.

9.2 Watermark removal for cross-posting

Important

TikTok applies a visible watermark to downloaded videos. YouTube and Instagram both detect this and reduce distribution. Remove it before cross-posting.

```
# FFmpeg: crop off TikTok watermark region (bottom-right corner)
# TikTok watermark is typically in bottom 80px of the 1080x1920 frame

ffmpeg -i tiktok_downloaded.mp4 \
  -vf "crop=1080:1840:0:0" \
  -c:a copy \
  clean_for_youtube.mp4

# Better approach: use the original MP4 from your R2 storage
# (before TikTok watermarks it) and upload that directly to each platform
# This is why storing your master MP4 in R2 before upload is critical
```

9.3 YouTube Data API upload

```
// YouTube Shorts upload via googleapis
const { google } = require('googleapis');
const youtube = google.youtube({ version: 'v3', auth: oauth2Client });

await youtube.videos.insert({
  part: ['snippet', 'status'],
```

```
requestBody: {
  snippet: {
    title: `${mangaTitle} Chapter ${chapterNum} Explained`,
    description: youtubeDescription, // longer than TikTok, include timestamps
    tags: ['manga', mangaTitle, 'shorts', 'anime', 'lore'],
    categoryId: '24', // Entertainment
  },
  status: {
    privacyStatus: 'public',
    selfDeclaredMadeForKids: false,
  }
},
media: { body: fs.createReadStream(videoFilePath) }
});
```

10. Cost model

10.1 Monthly cost breakdown — 150 videos/month

Service	Plan / usage	Per-unit cost	Monthly total
Claude API	~150 script gens	~\$0.05/script	~\$7.50
ElevenLabs	150 × 750 chars	\$0.30/1K chars	~\$33.75
Kling AI 2.1	450 clips (3/video)	~\$0.10/clip	~\$45.00
Creatomate	150 renders	~\$0.05/render	~\$7.50
Firecrawl	Starter plan	Flat rate	\$16.00
n8n self-hosted	VPS (2 core, 4GB)	Flat rate	~\$12.00
Cloudflare R2	150 × 50MB files	\$0.015/GB	~\$11.25
Perplexity API	~150 queries	~\$0.005/query	~\$0.75
Postgres (Supabase free tier)	< 500MB	Free	\$0.00
TOTAL	150 videos	~\$0.89/video	~\$133.75/mo

Revenue vs cost

At 150 videos/month with 3M total views (20K avg/video), Creator Rewards alone pays ~\$1,200–3,000. TikTok Shop affiliate at 0.1% conversion on 3M views and \$30 AOV at 12% commission = ~\$10,800/month. Total cost of \$134 against potential revenue of \$12,000+ is a 90x return on infrastructure spend.

11. Risk mitigation

11.1 Copyright and fair use

Risk level: HIGH

Using raw manga panels in monetized videos is the system's biggest legal exposure. Shueisha (JJK, One Piece), Kodansha (Attack on Titan, Fairy Tail), and Viz Media actively monitor TikTok and YouTube for unauthorized content.

Implementation requirements for fair use protection:

- Never show more than 3–4 panels per video — quantity signals reproduction, not commentary
- Every panel must be accompanied by original voiceover commentary in the script — transformation is the core fair use defense
- Use Kling AI to animate/stylize panels so they are derivative works, not direct reproductions
- For the safest long-term strategy: use Runway or Flux to generate original AI artwork inspired by the series visual style, replacing all copyrighted panel usage
- File DMCA counter-notices for wrongful strikes — document your commentary contribution in every video

11.2 Platform authenticity policies

TikTok Creator Rewards — inauthentic content exclusion

TikTok's 2025 policy explicitly excludes 'mass-produced content that lacks unique human contribution' from the Creator Rewards Program. Channels that appear fully automated are reviewed and can lose monetization access even after reaching thresholds.

Authenticity requirements to build into every script:

- Every hot-take hook must state a specific, debatable opinion — not a generic summary
- Reference the specific chapter by number in the video — proves current human engagement
- Mastra agent must generate a genuine opinion, not just summarize — tune the system prompt aggressively for this
- Vary the script format every 2–3 videos — never post the same structural template back-to-back

11.3 Account trust and ramp-up

Month 1

1–2 videos/day. Focus on building genuine engagement — respond to every comment manually. Do not post product ads yet.

Month 2	3 videos/day. Introduce product ads at 1:5 ratio (1 ad per 5 organic videos).
Month 3+	5+ videos/day. Apply for Creator Rewards once at 10K followers. Maintain 1:4 ad ratio maximum.
Multiple accounts	Run separate accounts per manga series — don't mix JJK and One Piece content on the same channel. Niche accounts outperform general ones on TikTok's recommendation algorithm.

11.4 Gameplay footage licensing

Subway Surfers	Kiloo has NOT issued a creator license. Avoid for monetized content. Use Pexels parkour or running footage instead — visually equivalent.
Minecraft	Microsoft's 'Content Usage Guidelines' permit creator use. Safe to use on monetized channels.
GTA V	Rockstar's policy does not explicitly permit monetized creator use. Use Pexels driving footage instead.
Best approach	Record your own gameplay sessions (30 min = 30+ days of B-roll). You own that footage outright with zero licensing risk.

12. Scaling roadmap

Phase 1 — Launch (Months 1–2)

- One manga series (JJK or One Piece — largest English-speaking fanbase)
- Manual quality review of every video before upload — build the pattern before automating approval
- 3 accounts: TikTok + YouTube Shorts + Instagram Reels
- Target: 10K TikTok followers and Creator Rewards eligibility
- Monthly infra cost: ~\$60 (50 videos/month)

Phase 2 — Expansion (Months 3–4)

- Add 2–3 manga series (Chainsaw Man, Berserk, Vinland Saga — high CPM audiences)
- Automate the quality approval step: Mastra scores final video before upload, rejects if below threshold
- Launch Whop community (\$9.99/month — early chapter summaries + exclusive tier lists)
- Add Amazon Associates + BookWalker affiliate links to YouTube video descriptions
- Monthly infra cost: ~\$134 (150 videos/month across 3 series)

Phase 3 — Monetization stack (Month 5+)

- YouTube long-form: weekly 10–15 minute 'full series breakdown' videos (3–5x higher CPM)
- Pinterest automation: Flux-generated art + reading guides driving passive Amazon affiliate income
- Brand partnerships: pitch anime merchandise brands directly (your media kit uses your TikTok analytics)
- Course on Whop or Kajabi: 'How to get into [series]' — use Claude to write the curriculum
- Monthly infra cost: ~\$200. Revenue potential: \$8,000–15,000/month from combined streams

The compound effect

The system's value compounds over time because: (1) Mastra learns which script styles work per series, (2) your channel accumulates SEO equity on YouTube search, (3) affiliate links in older videos continue earning without new effort, (4) Whop membership is recurring revenue — you keep earning from month-1 members forever. Build it once, optimize it monthly.